Mouhamed Amar

AI/ML Engineer | Reinforcement Learning & Robot Learning | Aspiring AI Research Scientist

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Research Interests

Recent M2 graduate in Artificial Intelligence and Big Data from ESP/UCAD, Senegal, with a research focus on **Reinforcement Learning, Robot Learning, and Embodied AI**. I am particularly interested in learning-based methods for robotic manipulation, sequential decision-making, and intelligent agents that perceive, reason, and act in physical environments. My interests span **reinforcement learning, imitation learning, vision-language-action (VLA) models, and learning-based robot control**.

I regularly **read and study research papers**, reproduce algorithms and experimental results, and build implementations from scratch to deepen my understanding of machine learning and robotics research. My recent work includes implementing RL algorithms from scratch and training imitation-learning policies such as **Action Chunking Transformer (ACT)** on ALOHA and **Behavior Cloning** on PushT.

Education

Master 2 – Intelligence Artificielle et Big Data (IABD) 2026
École Supérieure Polytechnique (ESP), Université Cheikh Anta Diop de Dakar (UCAD) *Dakar, Senegal*
- Thesis: *SDLC-AI – A Universal 11-Phase Engineering Framework for AI System Lifecycle Management*, research internship at UMMISCO
- Supervisors: Pr. Alassane Ba, Dr. Mandicou Ba, Dr. Fatou Ngom

Licence – Computer Science 2024
Université Cheikh Anta Diop de Dakar (UCAD) *Dakar, Senegal*

Research & Industry Experience

Research Intern – AI System Lifecycle Engineering 2025 – 2026
UMMISCO (Unité de Modélisation Mathématique et Informatique des Systèmes Complexes) *Dakar, Senegal*
- Designed and validated SDLC-AI, an 11-phase engineering framework for the AI system lifecycle
- Implemented a full MLOps pipeline (DVC, MLflow, SHAP, Evidently AI, Fairlearn, FastAPI, Airflow, GitHub Actions CI/CD) for cardiovascular disease prediction, validated on the UCI Heart Disease dataset – sdlcai-heart-disease
- Co-authored two scientific articles (state of the art and methodology) toward a doctoral research direction

AI Engineering Intern 2024 – 2025
SENELEC (Société Nationale d'Électricité du Sénégal) *Dakar, Senegal*
- Designed and developed a customer-facing RAG chatbot (FastAPI, Qdrant/LanceDB, BM25+FTS5, local LLM via Ollama) to automate and streamline client support requests – chatbot-api-senelec

AI Engineering Intern 2024
Ministère des Pêches et de l'Économie Maritime *Dakar, Senegal*
- Built a digital platform for the dematerialization of fishing license application and issuance processes

Selected Projects

LLM From Scratch – Architectures & Optimizers lm-from-scratch-architectures-and-optimizers
Built every major component of a modern LLM from scratch in PyTorch and benchmarked each variant experimentally (71 tests, 16 self-contained notebooks): tokenizers (char/byte-BPE, WordPiece, Unigram); positional encodings (sinusoidal, learned, RoPE with NTK/YaRN scaling, ALiBi, NoPE); attention (MHA, MQA, GQA, MLA, sliding window, QK-Norm); per-layer KV caching; normalization (LayerNorm, RMSNorm, Pre/Post-LN); FFN variants (GELU, SiLU, SwiGLU, GeGLU); Mixture-of-Experts (top-k routing, Switch/z-loss, bias-free routing, Expert Choice); and optimizers written by hand (SGD, Adam, AdamW, Lion, Muon) with WSD/cosine schedules, EMA, and exact checkpoint resumption.

Reinforcement Learning Algorithms from Scratch rl-algorithms
Implemented and compared core RL algorithms from scratch (PyTorch + Gymnasium): value/policy iteration (FrozenLake), Q-learning and SARSA (CliffWalking), DQN and Double DQN (CartPole), and REINFORCE (with/without baseline) and PPO (CartPole) – with controlled experiments isolating each algorithmic difference (e.g. off-policy vs. on-policy risk sensitivity, DQN overestimation bias, variance reduction from a REINFORCE baseline, PPO sample efficiency vs. Monte-Carlo policy gradient).

Mini RLHF Pipeline – SFT, Reward Model, PPO mini-RLHF-pipeline
Implemented the full three-stage InstructGPT-style RLHF pipeline from scratch (hand-written mini-GPT, no `transformers`):

supervised fine-tuning, a Bradley-Terry reward model, and PPO with a KL penalty against the reference policy. SFT loss 3.53 → 0.55; reward model preference accuracy ≈ 80%; PPO raised the sentiment reward from +0.2 to +5.1, while also surfacing and analyzing a concrete case of reward hacking.

LLM Supervised Fine-Tuning (SFT) Pipeline llm-sft-pipeline
Built an instruction-tuning pipeline for a base LLM (OLMo-2) with prompt masking (loss on response tokens only), gradient accumulation, linear warmup/decay scheduling, and full Weights & Biases experiment tracking.

Medical RAG Multi-Agent System (French) Assistant_Medical
Hybrid-retrieval multi-agent RAG system for French-language medical question answering: dense (LanceDB) + lexical (BM25) + graph (Neo4j) retrieval fused via Reciprocal Rank Fusion, orchestrated across five specialized agents (Gemini 2.0 Flash + Groq/Llama 3.3-70B & 3.1-8B), with a Streamlit interface and a rigorous evaluation notebook (ROUGE, BLEU, RAGAS); ROUGE-L ≈ 0.715.

Fine-Tuning Frontier Open Models (Qwen, Gemma) huggingface.co/Mouhamedamar
Fine-tuned frontier open-weight models (Qwen, Gemma) and SmolLM3-3B (QLoRA) for domain-specific tasks; contributed a pull request to Hugging Face's `smol-course` repository.

Imitation Learning: Action Chunking Transformer (ACT) – ALOHA action-chunking-transformer-Aloha
Trained an Action Chunking Transformer (ACT) policy for bimanual robot manipulation on the ALOHA dataset, using Hugging Face's `LeRobot` library for data loading and policy training; reproduced chunked-action prediction with temporal ensembling to reduce compounding error in behavior cloning.

Behavior Cloning from Scratch – PushT BC-pusht
Implemented a behavior cloning policy from scratch in a single PyTorch file for the PushT manipulation task, using `LeRobot`'s dataset tooling (`gym-pusht` / LeRobot PushT dataset) for data loading; designed to run end-to-end on CPU or in Colab.

Agentic VLM for Maritime Monitoring – Liquid AI Hackathon agentic-vlm-maritime-monitoring
Built an agentic vision-language system for maritime monitoring using Liquid AI's edge-optimized LFM2-VL model, developed during a Liquid AI (LEAP) hackathon.

Object Detection & Reasoning Pipelines
Implemented YOLO and R-CNN object detection pipelines from scratch; completed Hugging Face's Deep RL Course (PPO on LunarLander, MA-POCA on SoccerTwos).

Technical Skills

LLMs: Pretraining, fine-tuning (LoRA/QLoRA), supervised fine-tuning, quantization, KV cache optimization, tokenization, evaluation

Reinforcement Learning: Q-learning, DQN, policy gradients, REINFORCE, PPO, GRPO, RLOO, GSPO, CISPO, actor-critic methods

RLHF: Reward modeling, preference optimization, DPO, RLHF pipelines, RLVR (Reinforcement Learning with Verifiable Rewards)

Reasoning Models: Chain-of-thought, test-time compute, reasoning distillation, verifiable-reward training

Deep Learning & Machine Learning: PyTorch, CNNs, RNNs/Sequence models, Transformers, supervised & unsupervised learning, model evaluation & calibration

Agentic AI: Multi-agent systems, LangGraph, tool use, RAG (retrieval-augmented generation), agentic workflows

Robot Learning : Embodied AI, MuJoCo, Vision-Language-Action (VLA) models

MLOps: DVC, MLflow, SHAP, Evidently AI, Fairlearn, FastAPI, Airflow, GitHub Actions, Docker

Programming: Python, PyTorch, LaTeX, Git

Languages: French (native), Wolof (native), English (professional)

Certifications

- Introduction to Model Context Protocol – *Anthropic*
- Deep Reinforcement Learning Course – *Hugging Face*
- AI Agents Fundamentals – *Hugging Face*
- Fine-tuning Language Models – *Hugging Face*
- Supervised Finetuning – *Hugging Face*
- Natural Language Processing Specialization – *DeepLearning.AI*
- Deep Learning Specialization – *DeepLearning.AI*
- Machine Learning Specialization – *DeepLearning.AI*
- PyTorch for Deep Learning – *DeepLearning.AI*
- Convolutional Neural Networks – *DeepLearning.AI*
- Sequence Models – *DeepLearning.AI*
- Structuring Machine Learning Projects – *DeepLearning.AI*
- Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization – *DeepLearning.AI*
- Mathematics for Machine Learning and Data Science – *DeepLearning.AI*
- Linear Algebra for Machine Learning and Data Science – *DeepLearning.AI*
- Calculus for Machine Learning and Data Science – *DeepLearning.AI*
- Probability & Statistics for Machine Learning & Data Science – *DeepLearning.AI*

- AI for Everyone – *DeepLearning.AI*

Competitions & Community

- Zindi Telco Troubleshooting Agentic Challenge – Participant
- African Trust & Safety LLM Challenge – Participant (native Wolof/French fluency)
- Founder & lead contributor, French-language AI/ML technical education channel – technical syntheses of AI research (WhatsApp)
- Designed a 10-session AI club curriculum, from LLM foundations to research culture